

PARALLEL-PORT CONTROLLER USING MATLAB

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Many projects have been published in EFY on how to program a PC's parallel port using C, C++ and Visual Basic. Here we attempt the same using Matlab. The advantage of Matlab is that it does not require any special device driver to run in Windows XP. It has rich 'help' documents and a large number of user community in Internet, including Kluid and Matlab Central, which makes even an absolute begin-

ner create an application quickly.

Described here are two simple application programs in Matlab for controlling eight LEDs and a unipolar stepper motor through the parallel port. Since the files are bulky, you need a PC with at least 512MB RAM to run Matlab.

Parallel port

The parallel port of the PC is a 25-pin, D-type connector, commonly known as LPT (line printer terminal) port, having three different ports, namely, data port, control port and status port. Access to

the parallel port is via 25-pin, D-type female connector (DB25) as shown in Fig. 1. This female connector is normally available at the back side of your PC.

In order to access any port, you should have its port ID, i.e., the base address of the port. Here the data port has its base address as '0378' in hex format. The addresses of the three ports are in sequential order. So if the address of the data port is '0x378', the addresses of the status port and the control port will be '0x378+1' and '0x378+2', respectively.

Not all the 25 pins of the port are used for every application. Normally, the eight data lines (pins 2 through 9) along with the signal ground line are used. Pins 2

through 9 form the data port, thus making it an 8-bit wide port. It is a unidirectional port, i.e., it can be used only to output the data to the connected device. Pins 1, 14, 16 and 17 form the control port. It is a bidirectional port, i.e., it can be used both to input and output the data to and from a device. Pins 10 through 13 and 15 form the status port. It is a unidirectional port, i.e., it can be used only for data input from a device. The various pin functions of the parallel port are shown in Table I.

Circuit description

Fig. 2 shows the combined interfacing circuit for controlling the LEDs and the stepper motor through the parallel port. Here, LED1 through LED8 are connected to pins 2 through 9, respectively, of the parallel port. You can check these pins for proper working by clicking 'all on' and 'all off' buttons from the software (refer Fig. 3). All the eight LEDs will glow on clicking 'all on' button and turn off on clicking

PARTS LIST

Semiconductors:

- IC1 - 7812 12V regulator
- IC2-IC5 - MCT2E optocoupler
- IC6 - ULN2003 buffer/driver
- D1-D4 - 1N4001 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- LED1-LED9 - 5mm light-emitting diode
- R1 - 1-kilo-ohm
- R2-R9 - 220-ohm

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2 - 0.1 μ F ceramic

Miscellaneous:

- CON1 - 25-pin, D-type, parallel-port male connector
- X1 - 230V AC primary to 15V, 1A secondary transformer
- 555IM-25DAYG, 7.5 degrees/step, 12V stepper motor

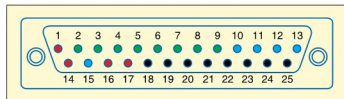


Fig. 1: 25-pin parallel-port connector (DB25)

TABLE I
Pin Details of 25-pin, D-type
Parallel-Port Connector

Pin	Register	Signal bit	Signal direction	Level	Hardware
1	Control	C 0	Output	Low	Inverted
2	Data	D 0	Output	High	Direct
3	Data	D 1	Output	High	Direct
4	Data	D 2	Output	High	Direct
5	Data	D 3	Output	High	Direct
6	Data	D 4	Output	High	Direct
7	Data	D 5	Output	High	Direct
8	Data	D 6	Output	High	Direct
9	Data	D 7	Output	High	Direct
10	Status	S 6	Input	High	Direct
11	Status	S 7	Input	Low	Inverted
12	Status	S 5	Input	High	Direct
13	Status	S 4	Input	High	Direct
14	Control	C 1	Output	Low	Inverted
15	Status	S 3	Input	High	Direct
16	Control	C 2	Output	High	Direct
17	Control	C 3	Output	Low	Inverted
18-25	Ground				

Note: Control register can be used as output as well as input